

MYP 1 – Mathematics

Unit 1: Numerical Foundations – Exam Practice Worksheet

Topics covered: Natural Numbers, Whole Numbers & Integers · Number Lines · Fractions & Decimals · Profit & Loss · Properties of Operations · Factors, Multiples, Prime Factorisation, LCM & HCF · BODMAS · Ratio & Proportion · Exponents, Squares & Cubes

Name: _____

Date: _____

Class: _____

Score: ____/ ____

Show all your working clearly. Write fractions in their simplest form unless told otherwise. A full answer key is provided at the end of this worksheet.

Section A Natural Numbers, Whole Numbers & Integers

A1. Classify each number by placing a tick (✓) in the correct column(s):

Number	Natural	Whole	Integer	Rational
-7				
0				
5				
-3/4				
12				
-100				

A2. Write the following sets in ascending order (smallest to largest):

a) $\{-5, 3, -8, 0, 2, -1\}$	b) $\{7, -7, 4, -4, 0, -2\}$
c) $\{-100, -50, 25, -75, 0\}$	d) $\{-3, 1/2, -1, 2, -2, 0\}$

Section B Integers on the Number Line

B1. Draw a number line from -10 to $+10$ and mark the following points:

$-8, -5, -1, 0, 3, 6, 9$

B2. Use the number line to find the distance between each pair of integers:

a) -3 and 5	b) -7 and -2	c) 4 and -6
d) 0 and -9	e) -4 and 4	f) -10 and 3

B3. Write the integer that is:

a) 4 more than -3	b) 6 less than 2	c) 7 less than -1
d) 5 more than -8	e) halfway between -6 and 4	f) halfway between -9 and -1

B4. Compare each pair using $>$ or $<$ or $=$:

a) -3 ____ 2	b) -5 ____ -8	c) $ -4 $ ____ $ 3 $
d) 0 ____ -1	e) -7 ____ -7	f) $ -6 $ ____ $ 6 $

Section C Fractions & Decimals

C1. Convert each fraction to a decimal. State whether it is Terminating (T) or Repeating (R):

a) $\frac{3}{4}$	b) $\frac{5}{6}$	c) $\frac{7}{8}$
d) $\frac{2}{3}$	e) $\frac{5}{11}$	f) $\frac{9}{20}$

g) $\frac{4}{9}$	h) $\frac{7}{12}$	i) $\frac{11}{16}$
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C2. Convert each decimal to a fraction in simplest form:

a) 0.8	b) 0.35	c) 0.625
d) 0.125	e) 0.45	f) 1.75

C3. Compare each pair using >, < or =:

a) $\frac{3}{5}$ ____ 0.62	b) $\frac{7}{8}$ ____ 0.875	c) $\frac{2}{3}$ ____ 0.65
d) $\frac{5}{4}$ ____ 1.26	e) 0.6 ____ $\frac{2}{3}$	f) 0.37 ____ $\frac{3}{8}$

Section D Basic Profit & Loss

Reminder: Profit = Selling Price – Cost Price | Loss = Cost Price – Selling Price | Profit% or Loss% is always calculated on Cost Price

D1. State whether there is a profit or loss, and calculate the amount:

Item	Cost Price (CP)	Selling Price (SP)	Profit or Loss?	Amount (₹)
Book	₹80	₹100		
Bag	₹450	₹390		
Bicycle	₹3,200	₹4,000		
Watch	₹1,500	₹1,200		
Shirt	₹600	₹720		

D2. Find the missing value:

#	Cost Price (CP)	Selling Price (SP)	Profit / Loss	Working
a)	₹200	?	Profit of ₹50	
b)	?	₹840	Loss of ₹60	

c)	₹950	₹1,140	?	
d)	₹2,400	?	Loss of ₹300	

D3. Word Problems:

a) Arjun bought a second-hand phone for ₹8,000 and sold it for ₹6,500. Find his loss.

b) A shopkeeper bought 20 notebooks at ₹15 each and sold all of them for ₹360 total. Did he make a profit or a loss? By how much?

c) Priya bought a dress for ₹1,200 and sold it at a profit of ₹180. What was the selling price?

d) ★ A trader bought apples for ₹5 per kg and sold them for ₹7 per kg. If he sold 40 kg, find the total profit.

Section E Properties of Operations (Commutative, Associative, Distributive, Identity)

E1. Name the property shown (Commutative / Associative / Distributive / Identity):

a) $5 + 9 = 9 + 5 \rightarrow$ _____	b) $(3 \times 4) \times 7 = 3 \times (4 \times 7) \rightarrow$ _____	c) $8 \times (5 + 2) = 8 \times 5 + 8 \times 2 \rightarrow$ _____
d) $-6 + 0 = -6 \rightarrow$ _____	e) $(-4 + 7) + 3 = -4 + (7 + 3) \rightarrow$ _____	f) $12 \times (10 - 3) = 12 \times 10 - 12 \times 3 \rightarrow$ _____
g) $1 \times 15 = 15 \rightarrow$ _____	h) $23 + 41 = 41 + 23 \rightarrow$ _____	i) $7 \times 1 = 7 \rightarrow$ _____

E2. Use the distributive property to calculate mentally. Show your method:

a) $6 \times (20 + 3)$	b) 9×98
c) 15×102	d) 7×49

e) $25 \times (40 - 1)$

f) 8×53

E3. Fill in the missing number and name the property used:

a) $7 + \underline{\quad} = 4 + 7$

b) $(\underline{\quad} \times 3) \times 6 = 5 \times (3 \times 6)$

c) $9 \times (4 + 2) = 9 \times 4 + 9 \times \underline{\quad}$

d) $\underline{\quad} + 0 = -13$

e) $1 \times \underline{\quad} = 27$

f) $(2+8) + \underline{\quad} = 2 + (8+6)$

Section F Factors, Multiples & Prime Factorisation**F1. List all the factors of each number:**

a) 18

b) 24

c) 36

d) 48

e) 72

f) 100

F2. List the first 6 multiples of each number:

a) 4

b) 7

c) 9

d) 12

e) 15

f) 20

F3. State Prime or Composite. If composite, write its prime factorisation:

a) 17

b) 28

c) 37

d) 49

e) 53

f) 77

g) 91

h) 83

i) 99

F4. Write each number as a product of prime factors in index form:

a) 36	b) 60	c) 72
d) 120	e) 180	f) 252

Section G LCM & HCF of Two Numbers**G1. Find the HCF and LCM of each pair using prime factorisation. Show your working:**

a) 12 and 18	b) 24 and 36
c) 15 and 35	d) 30 and 45
e) 16 and 40	f) 21 and 28
g) 48 and 64	h) 18 and 27

G2. Word Problems:

a) Two bells ring at intervals of 12 minutes and 18 minutes. If they ring together at 10:00 a.m., when will they next ring together?

b) Riya has 36 red pens and 48 blue pens. She wants to pack them into identical bags with the same number of each colour in every bag, with none left over. What is the greatest number of bags she can make? How many of each pen per bag?

c) Two ropes of lengths 42 cm and 56 cm are cut into equal pieces of the greatest possible length with no waste. What is the length of each piece? How many pieces in total?

d) ★ The HCF of two numbers is 8 and their LCM is 96. If one number is 24, find the other.

Section H BODMAS Rule with Integers

H1. Evaluate (integer operations):

a) $-6 + (+9)$	b) $-4 - 7$	c) $8 - (+11)$
d) $5 - (-3)$	e) $-8 - (+6)$	f) $-10 - (-4)$
g) $3 - 14$	h) $-7 + 12$	i) $15 - 22$
j) $-20 - 15$	k) $(-5) + (+8)$	l) $11 - (-9)$

H2. Evaluate using the correct order of operations:

a) $(-10) + (-8) \div (-4)$	b) $(-15) + (+12)(-2)$
c) $(-6 \times 3) - (-18) \div 3$	d) $(-7 - 3)(-5)$
e) $(+8) - (-2)^2$	f) $16 \div (-8) + (-3)$

H3. Evaluate these longer expressions:

a) $(+12) + (-8) + (+5) + (-6)$	b) $(-5) - (+10) + (-4) - (-7)$
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c) $(-64) \div (+8)^2 + (-100) \div (-5)^2$	d) $(+8)(-9) \div (-3)^2$
e) $6 - (-36) \div (+6) + (-4)(-5) - (+9)$	f) $[(+18) - (-42)] \div [(-6)(3) - (-2)]$

Section I Addition, Subtraction & Simplification of Fractions and Decimals

I1. Add or subtract. Give your answer in simplest form:

a) $1/2 + 1/4$	b) $3/5 + 2/3$	c) $7/8 - 1/4$
d) $5/6 - 1/3$	e) $2 \frac{1}{3} + 1 \frac{1}{4}$	f) $4 \frac{1}{2} - 2 \frac{3}{4}$
g) $3/4 + 5/6$	h) $7/10 - 2/5$	i) $1 \frac{5}{6} + 2 \frac{3}{4}$

I2. Add or subtract these decimals:

a) $3.6 + 2.85$	b) $7.04 - 1.8$	c) $0.75 + 1.4 + 0.05$
d) $5.3 - 2.075$	e) $12.6 + 0.045$	f) $9 - 3.27$

I3. Simplify each fraction fully:

a) $18/24$	b) $35/49$	c) $60/80$
d) $72/108$	e) $45/75$	f) $84/126$

I4. Convert between mixed numbers and improper fractions:

a) $3 \frac{2}{5} \rightarrow$	b) $7/4 \rightarrow$	c) $2 \frac{3}{8} \rightarrow$
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d) $11/3 \rightarrow$ e) $5 \frac{1}{6} \rightarrow$ f) $19/6 \rightarrow$

Section J Introduction to Ratio & Proportion

J1. Simplify each ratio:

a) $6 : 10$ b) $15 : 25$ c) $24 : 36$ d) $0.4 : 2$ e) $1.5 : 4.5$ f) $1\frac{1}{2} : 2\frac{1}{2}$

J2. Find the missing value so the ratios are equivalent:

a) $2 : 5 = x : 20$ b) $3 : 7 = 9 : y$ c) $5 : 8 = x : 40$ d) $a : 15 = 4 : 5$ e) $6 : b = 18 : 27$ f) $7 : 3 = 35 : c$

J3. Divide each amount in the given ratio:

a) Divide ■400 in the ratio $3 : 5$ b) Divide 630 g in the ratio $2 : 7$ c) Divide 480 in the ratio $1 : 2 : 3$ d) Divide ■1,200 in the ratio $5 : 3 : 4$

J4. Word Problems:

a) In a class of 35 students, the ratio of boys to girls is $3 : 4$. How many boys and how many girls are in the class?

b) A fruit juice is made by mixing mango and pineapple in the ratio $3 : 2$. If 12 litres of mango juice are used, how much pineapple juice is needed? What is the total volume?

c) The angles of a triangle are in the ratio $1 : 2 : 3$. Find each angle. What type of triangle is it?

d) If 6 pencils cost ₹18, find the cost of 10 pencils using the unitary method.

Section K Exponent Notation, Square Numbers & Cubes

K1. Write in exponent notation:

a) $5 \times 5 \times 5$	b) $3 \times 3 \times 3 \times 3$	c) $2 \times 2 \times 2 \times 2 \times 2$
d) $10 \times 10 \times 10 \times 10$	e) 7×7	f) $6 \times 6 \times 6 \times 6 \times 6$

K3. Evaluate:

a) 2^4	b) 3^3	c) 4^2
d) 2^5	e) 5^3	f) 10^3
g) 1^{10}	h) 6^2	i) 7^2

K4. Complete the square and cube number tables:

Number (n)	1	2	3	4	5	6	7	8	9	10
n^2 (Square)										
n^3 (Cube)										

K5. State whether each number is a perfect square, perfect cube, both, or neither:

a) 25	b) 27	c) 64
d) 36	e) 100	f) 125
g) 144	h) 8	i) 50

ANSWER KEY

Section A – Natural Numbers, Whole Numbers & Integers

A1. -7: Integer, Rational ✓ | 0: Whole, Integer, Rational ✓ | 5: Natural, Whole, Integer, Rational ✓ | -3/4: Rational ✓ | 12: Natural, Whole, Integer, Rational ✓ | -100: Integer, Rational ✓

A2. a) -8, -5, -1, 0, 2, 3 b) -7, -4, 0, 4, 7 (Note: 7 last) c) -100, -75, -50, 0, 25 d) -2, -1, 0, 1/2, 2

A4. a) 8 b) 2 c) 16 d) 3 e) 12 f) 5

Section B – Integers on the Number Line

B2. a) 8 b) 5 c) 10 d) 9 e) 8 f) 13

B3. a) 1 b) -4 c) -8 d) -3 e) -1 f) -5

B4. a) < b) > c) > d) > e) = f) =

Section C – Fractions & Decimals

C1. a) 0.75 (T) b) 0.8 ■ 3 ■ (R) c) 0.875 (T) d) 0.6 ■ (R) e) 0.4 ■ 5 ■ (R) f) 0.45 (T) g) 0.4 ■ (R) h) 0.583 ■ (R) i) 0.6875 (T)

C2. a) 4/5 b) 7/20 c) 5/8 d) 1/8 e) 9/20 f) 7/4 (1¾)

C4. a) < b) = c) > d) < e) = f) >

Section D – Profit & Loss

D1. Book: Profit ₹20 | Bag: Loss ₹60 | Bicycle: Profit ₹800 | Watch: Loss ₹300 | Shirt: Profit ₹120

D2. a) SP = ₹250 b) CP = ₹900 c) Profit = ₹190 d) SP = ₹2,100

D3. a) Loss = ₹1,500 b) SP = ₹360, CP = ₹300 → Profit of ₹60 c) SP = ₹1,380 d) ★ Profit = (7-5)×40 = ₹80

Section E – Properties of Operations

E1. a) Commutative b) Associative c) Distributive d) Identity (addition) e) Associative f) Distributive g) Identity (multiplication) h) Commutative i) Identity (multiplication)

E2. a) $6 \times 20 + 6 \times 3 = 138$ b) $9 \times 100 - 9 \times 2 = 882$ c) $15 \times 100 + 15 \times 2 = 1,530$ d) $7 \times 50 - 7 \times 1 = 343$ e) $25 \times 40 - 25 \times 1 = 975$ f) $8 \times 50 + 8 \times 3 = 424$

E3. a) 4, Commutative b) 5, Associative c) 2, Distributive d) -13, Identity e) 27, Identity f) 6, Associative

Section F – Factors, Multiples & Prime Factorisation

F1. a) 1,2,3,6,9,18 b) 1,2,3,4,6,8,12,24 c) 1,2,3,4,6,9,12,18,36 d) 1,2,3,4,6,8,12,16,24,48 e) 1,2,3,4,6,8,9,12,18,24,36,72 f) 1,2,4,5,10,20,25,50,100

F2. a) 4,8,12,16,20,24 b) 7,14,21,28,35,42 c) 9,18,27,36,45,54 d) 12,24,36,48,60,72 e) 15,30,45,60,75,90 f) 20,40,60,80,100,120

F3. a) 17 Prime b) $28 = 2^2 \times 7$ Composite c) 37 Prime d) $49 = 7^2$ Composite e) 53 Prime f) $77 = 7 \times 11$ Composite g) $91 = 7 \times 13$ Composite h) 83 Prime i) $99 = 3^2 \times 11$ Composite

F4. a) $2^2 \times 3^2$ b) $2^2 \times 3 \times 5$ c) $2^3 \times 3^2$ d) $2^3 \times 3 \times 5$ e) $2^2 \times 3^2 \times 5$ f) $2^2 \times 3^2 \times 7$

Section G – LCM & HCF

G1. a) HCF 6, LCM 36 b) HCF 12, LCM 72 c) HCF 5, LCM 105 d) HCF 15, LCM 90 e) HCF 8, LCM 80 f) HCF 7, LCM 84 g) HCF 16, LCM 192 h) HCF 9, LCM 54

G2. a) $\text{LCM}(12,18)=36$ min → 10:36 a.m. b) $\text{HCF}(36,48)=12$ bags; 3 red, 4 blue per bag c) $\text{HCF}(42,56)=14$ cm; $3+4=7$ pieces d) ★ $\text{HCF} \times \text{LCM} = 4 \times 48 = 192$; other = $192 \div 24 = 8$

Section H – BODMAS with Integers

H1. a) 3 b) -11 c) -3 d) 8 e) -14 f) -6 g) -11 h) 5 i) -7 j) -35 k) 3 l) 20

H2. a) -8 b) -39 c) -12 d) 50 e) 4 f) -5

H3. a) 3 b) -12 c) -2 d) -8 e) 17 f) $-20/3$

Section I – Fractions & Decimals — Operations

I1. a) $3/4$ b) $19/15$ (1 $4/15$) c) $5/8$ d) $1/2$ e) $3\ 7/12$ f) $1\ 3/4$ g) $19/12$ (1 $7/12$) h) $3/10$ i) $4\ 7/12$

I2. a) 6.45 b) 5.24 c) 2.2 d) 3.225 e) 12.645 f) 5.73

I3. a) $3/4$ b) $5/7$ c) $3/4$ d) $2/3$ e) $3/5$ f) $2/3$

I4. a) $17/5$ b) $1\frac{3}{4}$ c) $19/8$ d) $3\frac{1}{2}$ e) $31/6$ f) $3\frac{1}{2}$

Section J – Ratio & Proportion

J1. a) 3:5 b) 3:5 c) 2:3 d) 1:5 e) 1:3 f) 3:5

J2. a) $x=8$ b) $y=21$ c) $x=25$ d) $a=12$ e) $b=9$ f) $c=15$

J3. a) $\blacksquare 150$, $\blacksquare 250$ b) 140 g, 490 g c) 80, 160, 240 d) $\blacksquare 500$, $\blacksquare 300$, $\blacksquare 400$

J4. a) 15 boys, 20 girls b) 8 L pineapple, 20 L total c) 30° , 60° , 90° — right-angled triangle d) $\blacksquare 3$ per pencil $\rightarrow$ 10 pencils = $\blacksquare 30$

Section K – Exponents, Squares & Cubes

K1. a) 5^3 b) $3\frac{1}{2}$ c) $2\frac{1}{2}$ d) $10\frac{1}{2}$ e) 7^2 f) $6\frac{1}{2}$

K3. a) 16 b) 27 c) 16 d) 32 e) 125 f) 1,000 g) 1 h) 36 i) 49

K4. Squares: 1,4,9,16,25,36,49,64,81,100 | Cubes: 1,8,27,64,125,216,343,512,729,1000

K5. a) Perfect square (5^2) b) Perfect cube (3^3) c) Both ($8=2^3$, $64=8^2=4^3$) d) Perfect square (6^2) e) Perfect square (10^2) f) Perfect cube (5^3) g) Perfect square (12^2) h) Perfect cube (2^3) i) Neither

End of Worksheet | MYP 1 Mathematics – Unit 1: Numerical Foundations